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MICROPROCESSOR AND INTERFACING

(3160712)

Winter – 2024 – Solution

Question : 01

Question 1(a)

List the key parts of a microprocessor and give a short explanation of their functions.

Answer:

The main parts of a microprocessor are explained below:

1. Arithmetic and Logic Unit (ALU)

- Performs all **arithmetic operations** like addition and subtraction.
- Performs **logical operations** like AND, OR, NOT, XOR.
- It is the main calculating unit of the microprocessor.

2. Control Unit (CU)

- Controls all operations inside the microprocessor.
- Gives control signals to different units.
- Manages the execution of instructions step-by-step.

3. Registers

- Small and fast memory locations inside the CPU.
- Used to store temporary data and instructions.

Types of Registers:

- **Accumulator (A):** Stores arithmetic and logical results.
- **Program Counter (PC):** Stores address of next instruction.

- **Stack Pointer (SP):** Points to top of stack memory.
- **Flag Register:** Stores status flags like Zero, Carry, Sign, etc.

4. Bus System

Used for communication between CPU, memory, and I/O devices.

Types of Buses:

- **Address Bus:** Carries address information.
- **Data Bus:** Transfers data.
- **Control Bus:** Carries control signals.

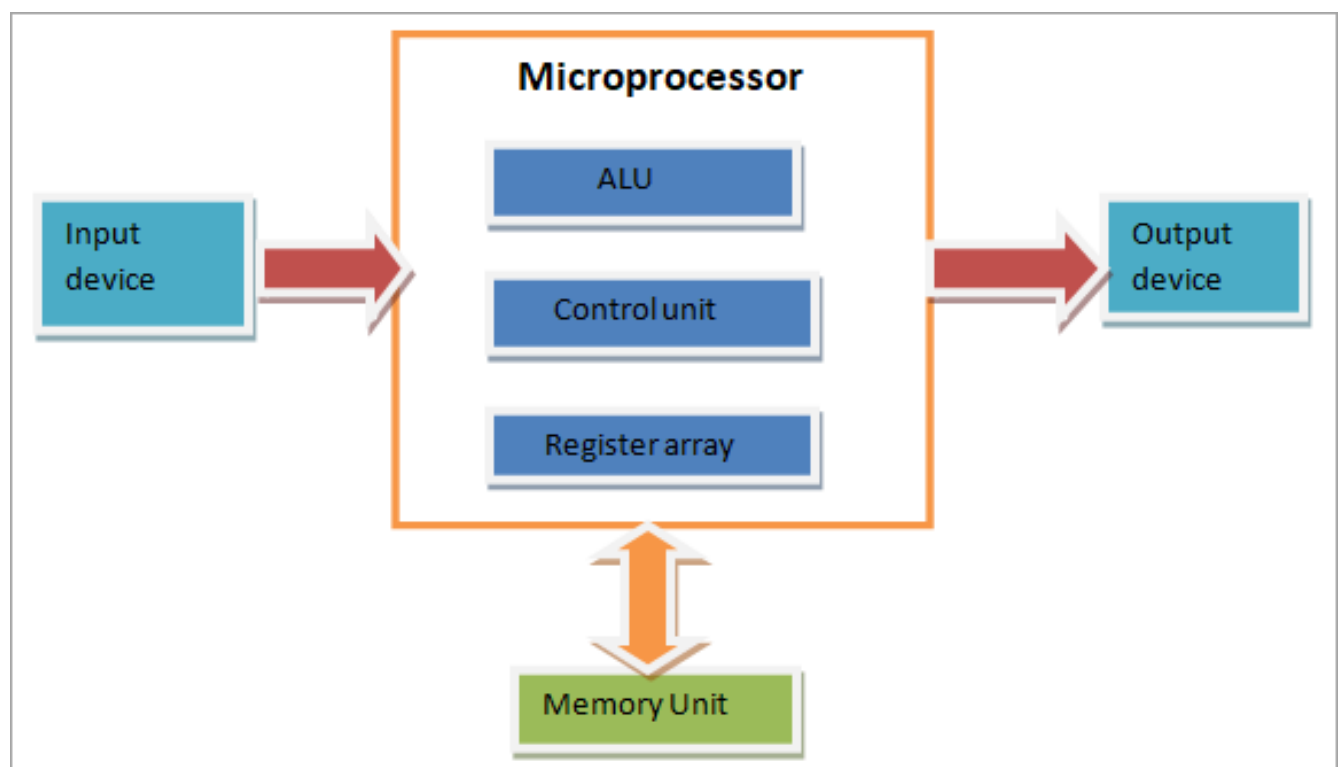
5. Timing and Control Unit

- Generates timing signals.
- Synchronizes all operations of the microprocessor.

6. Memory Unit

- Stores program instructions and data.
- Includes RAM and ROM.

Block Diagram of Microprocessor



Conclusion

A microprocessor works by using ALU, control unit, registers, buses, and memory together to execute instructions and process data efficiently.

Question 1(b)

Explain memory and I/O operations in a microprocessor.

Answer:

A microprocessor communicates with **memory** and **input/output devices** to perform operations.

1. Memory Operations

Memory operations are used to **read data from memory** or **write data into memory**.

Types of Memory Operations

(a) Memory Read Operation

- Microprocessor reads data from memory.
- The address of memory location is placed on the address bus.
- Control signal **MEM $\bar{R}$** becomes active.
- Data is transferred from memory to microprocessor.

Example:

LDA 2050H

This instruction reads data from memory location 2050H.

(b) Memory Write Operation

- Microprocessor writes data into memory.
- Address is sent through address bus.
- Control signal **MEM $\bar{W}$** becomes active.
- Data is transferred from microprocessor to memory.

Example:

STA 3050H

This instruction stores accumulator data into memory location 3050H.

2. I/O Operations

I/O operations are used to exchange data with input and output devices.

Types of I/O Operations

(a) I/O Read Operation

- Microprocessor reads data from an input device.
- Control signal $\text{IO}\bar{\text{R}}$ becomes active.

Example:

IN 20H

Reads data from input port 20H.

(b) I/O Write Operation

- Microprocessor sends data to output device.
- Control signal $\text{IO}\bar{\text{W}}$ becomes active.

Example:

OUT 30H

Sends data to output port 30H.

Difference Between Memory and I/O Operations

Memory Operation	I/O Operation
Accesses memory locations	Accesses input/output devices
Uses $\text{MEM}\bar{\text{R}}$ and $\text{MEM}\bar{\text{W}}$ signals	Uses $\text{IO}\bar{\text{R}}$ and $\text{IO}\bar{\text{W}}$ signals
Large address space	Smaller address space
Stores programs and data	Transfers external data

Conclusion

Memory operations handle data storage and retrieval, while I/O operations manage communication between the microprocessor and external devices.

Question 1(c)

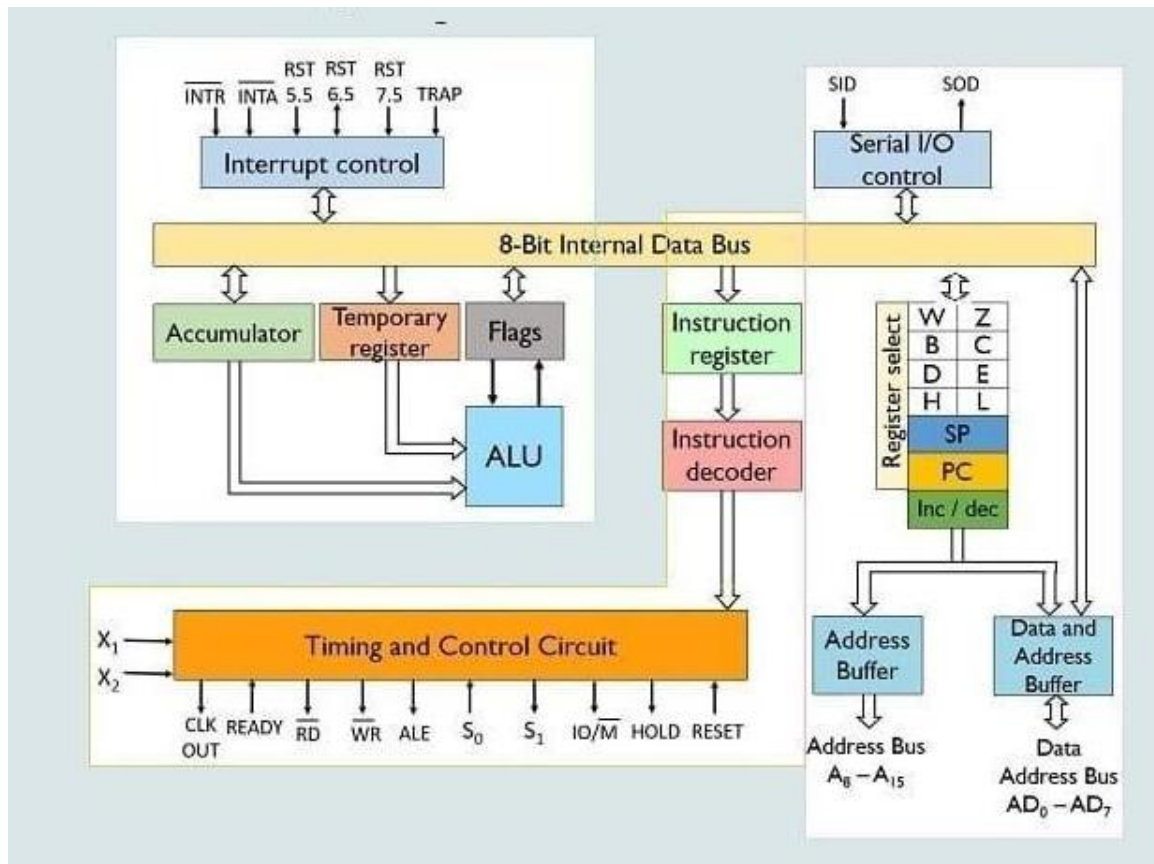
Describe the architecture of the 8085 microprocessor with a neat diagram.

Answer:

The **8085 microprocessor** is an 8-bit microprocessor developed by Intel. It has:

- **8-bit data bus**
- **16-bit address bus**
- **Can access 64 KB memory**

Block Diagram of 8085 Microprocessor



Main Components of 8085 Architecture

1. Arithmetic and Logic Unit (ALU)

- Performs arithmetic and logical operations.
- Works with accumulator.

Operations:

- Addition
- Subtraction
- AND, OR, XOR
- Compare

2. Accumulator

- 8-bit register.
- Stores result of ALU operations.

3. Register Array

Contains six 8-bit general purpose registers:

- B
- C
- D
- E
- H
- L

These registers can also work as register pairs:

- BC
- DE
- HL

4. Flag Register

Stores status of operations.

Flags in 8085:

- Sign Flag (S)
- Zero Flag (Z)
- Auxiliary Carry (AC)
- Parity Flag (P)
- Carry Flag (CY)

5. Program Counter (PC)

- 16-bit register.
- Stores address of next instruction.

6. Stack Pointer (SP)

- 16-bit register.
- Points to top of stack memory.

7. Instruction Register and Decoder

- Instruction register stores current instruction.
- Decoder decodes instruction for execution.

8. Timing and Control Unit

- Generates timing and control signals.
- Controls all internal and external operations.

9. Interrupt Control

Handles interrupts like:

- TRAP
- RST 7.5
- RST 6.5
- RST 5.5
- INTR

10. Serial I/O Control

Used for serial communication using:

- SID (Serial Input Data)
- SOD (Serial Output Data)

Features of 8085

- 8-bit microprocessor
- Operates at 3 MHz clock frequency
- 16-bit address bus
- 8-bit data bus
- Supports interrupts
- Single +5V power supply

Conclusion

The 8085 architecture consists of ALU, registers, control unit, buses, and interrupt system working together to execute instructions and process data efficiently.